Akoizo Original Practice - 2027 Rotation Form 2

Division B | Water Quality | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Explain why this transect is an estuary rather than a freshwater-only habitat.

Synthetic estuary transect: river end salinity 0, marine end salinity 24 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.4 to 4.4 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.6 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
_____
_____

2. (2 points) Explain how a nutrient pulse can lead to lower overnight oxygen.

Synthetic estuary transect: river end salinity 0, marine end salinity 24 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.4 to 4.4 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.6 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
_____
_____

3. (2 points) Identify one watershed action that could reduce nutrient loading and explain its pathway.

Synthetic estuary transect: river end salinity 0, marine end salinity 24 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.4 to 4.4 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.6 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
_____
_____

4. (2 points) Explain how traditional ecological knowledge could improve this monitoring plan without replacing measured evidence.

Synthetic estuary transect: river end salinity 0, marine end salinity 24 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.4 to 4.4 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.6 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
_____
_____

5. (2 points) Why should a reef food-web analysis distinguish herbivores from predators?

Answer: _____
_____
_____

6. (2 points) Distinguish a harmful algal bloom from an ordinary observation of green water.

Answer: _____
_____
_____

7. (2 points) A dark urchin has conspicuously long slender spines and grazes algae. Identify its listed group.

Answer: _____
_____
_____

8. (2 points) A shrimp has red-and-white bands and conspicuous elongated appendages. Identify the listed common-name group.

Answer: _____
_____
_____

9. (2 points) An elongate, soft-bodied echinoderm processes sediment on the seafloor. Identify its listed group.

Answer: _____
_____
_____

10. (2 points) Name the listed Indo-Pacific bivalve group with a large shell, mantle and photosynthetic symbionts.

Answer: _____
_____
_____

11. (2 points) An elongate predatory fish shelters in reef crevices and has strong jaws. Identify the listed eel group.

Answer: _____
_____
_____

12. (2 points) Name the listed overgrowing non-coral macroalgal category that can compete for reef space.

Answer: _____
_____
_____

13. (2 points) Calculate the overnight dissolved-oxygen decline.

Synthetic estuary transect: river end salinity 0, marine end salinity 24 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.4 to 4.4 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.6 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
_____
_____

14. (2 points) Calculate the BOD-bottle oxygen demand under the stated assumptions.

Synthetic estuary transect: river end salinity 0, marine end salinity 24 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.4 to 4.4 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.6 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
_____
_____

15. (2 points) A mixture contains equal volumes of freshwater and seawater with salinity 24. Assuming additive volumes and conservative mixing, estimate salinity.

Synthetic estuary transect: river end salinity 0, marine end salinity 24 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.4 to 4.4 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.6 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
_____
_____

16. (2 points) Aragonite saturation decreases while other factors are held fixed. Explain a relevant concern for stony corals.

Answer: _____
_____
_____

17. (2 points) Explain why high turbidity and high dissolved-solids concentration are not interchangeable observations.

Answer: _____
_____
_____

18. (2 points) A field pH value changes between daytime and predawn. Name a biological mechanism consistent with this and one check on the inference.

Answer: _____
_____
_____

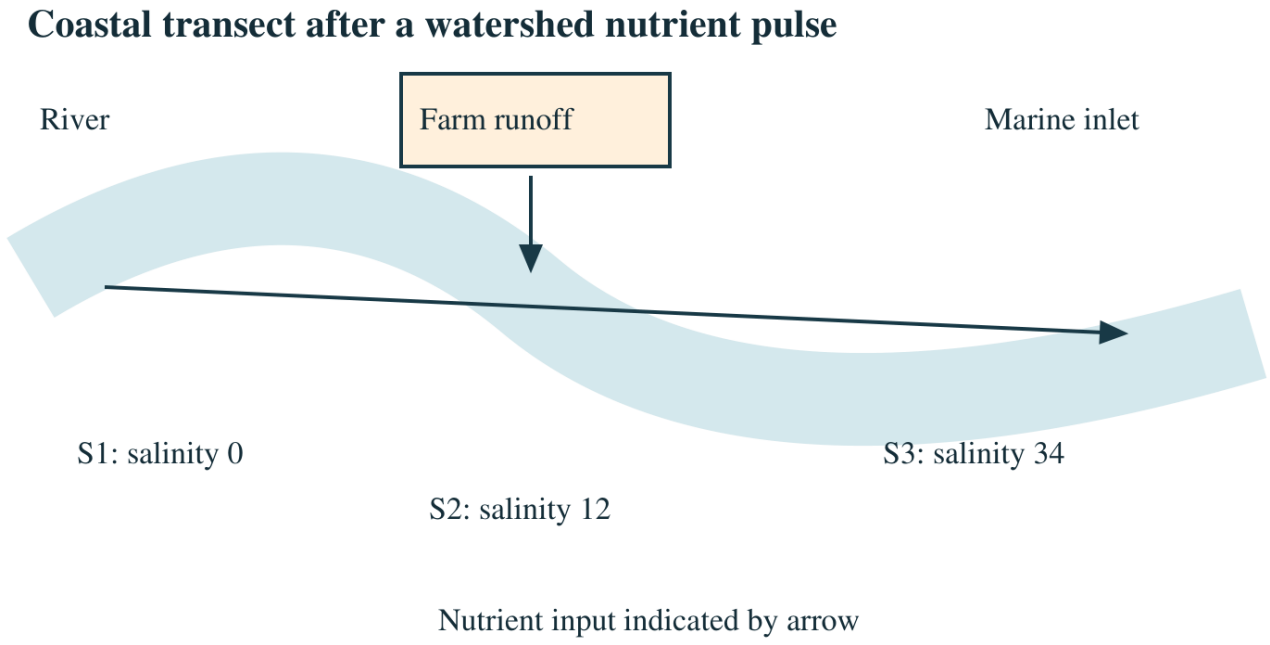
19. (2 points) A salinometer calibration gives reading 10 at 2% mass/volume salt and reading 30 at 6%. A sample reads 17. Use linear interpolation to estimate concentration.

Answer: _____
_____
_____

20. (2 points) How many grams of salt are required for 600 mL of final solution at 3% mass/volume?

Answer: _____
_____
_____

21. (4 points) Use Figure F1. Which labeled station most clearly represents brackish mixing rather than either freshwater or full seawater? Explain.



F1. Estuary sampling map (original synthetic teaching figure)

Answer: _____
_____
_____

22. (4 points) Use Figure F1. Trace a plausible route from the mapped nutrient source to reef oxygen stress.

Refer to F1. Estuary sampling map (original synthetic teaching figure) shown with the first question in this figure group.

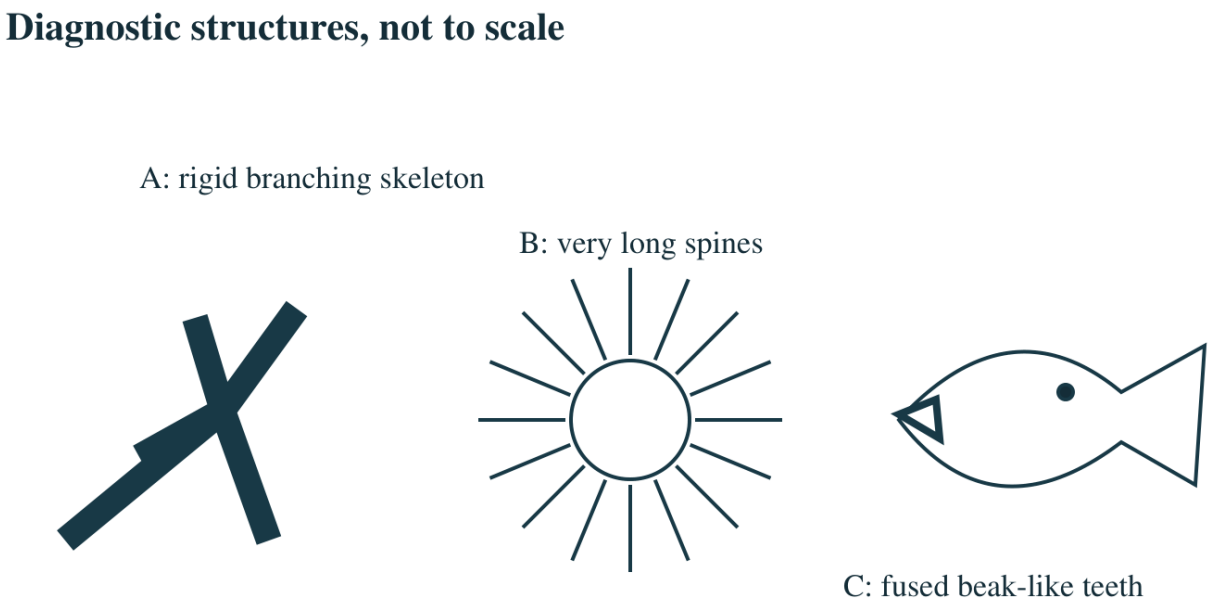
Answer: _____
_____
_____

23. (4 points) Use Figure F1. Place a vegetated buffer at an appropriate location on the map and explain its purpose.

Refer to F1. Estuary sampling map (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (4 points) Use Figure F2. Identify the listed reef-building group represented by A.



F2. Reef organism identification sketches (original synthetic teaching figure)

Answer: _____
_____
_____

25. (4 points) Use Figure F2. Identify the listed urchin group represented by B.

Refer to F2. Reef organism identification sketches (original synthetic teaching figure) shown with the first question in this figure group.

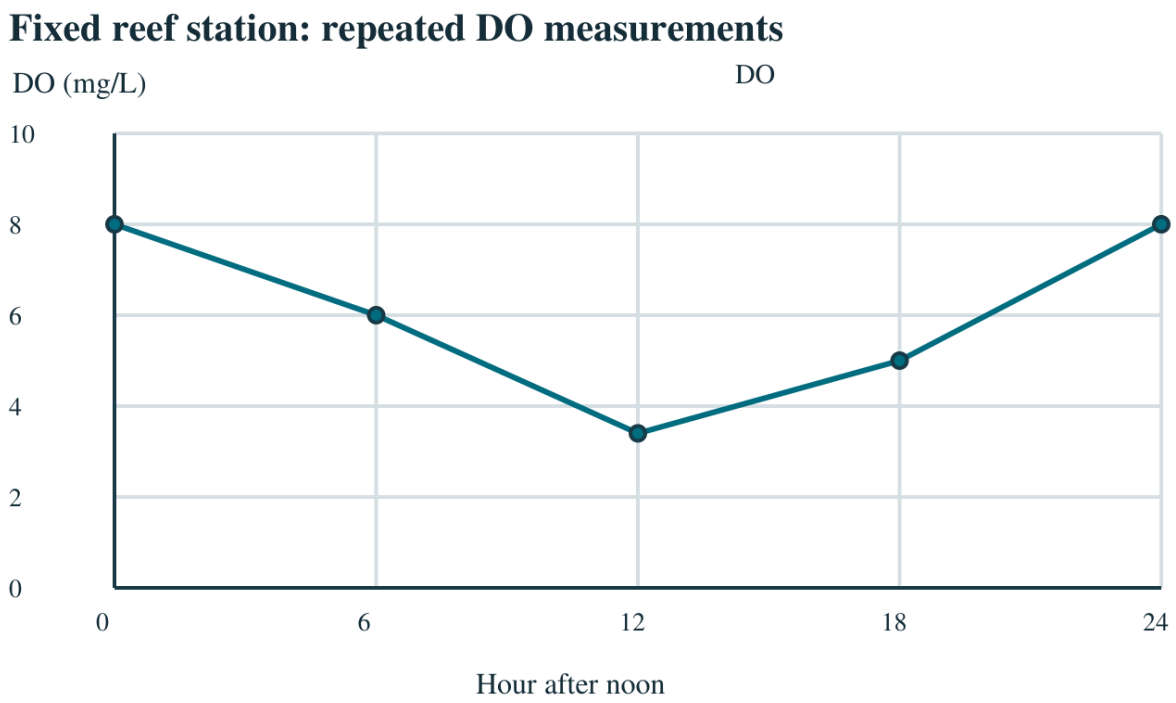
Answer: _____
_____
_____

26. (4 points) Use Figure F2. Identify C and explain how its feeding could affect competition for reef space.

Refer to F2. Reef organism identification sketches (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (4 points) Use Figure F3. Read the lowest observed DO concentration.



F3. Diel dissolved oxygen (original synthetic teaching figure)

Answer: _____
_____
_____

28. (4 points) Use Figure F3. Calculate the observed maximum-to-minimum DO range.

Refer to F3. Diel dissolved oxygen (original synthetic teaching figure) shown with the first question in this figure group.

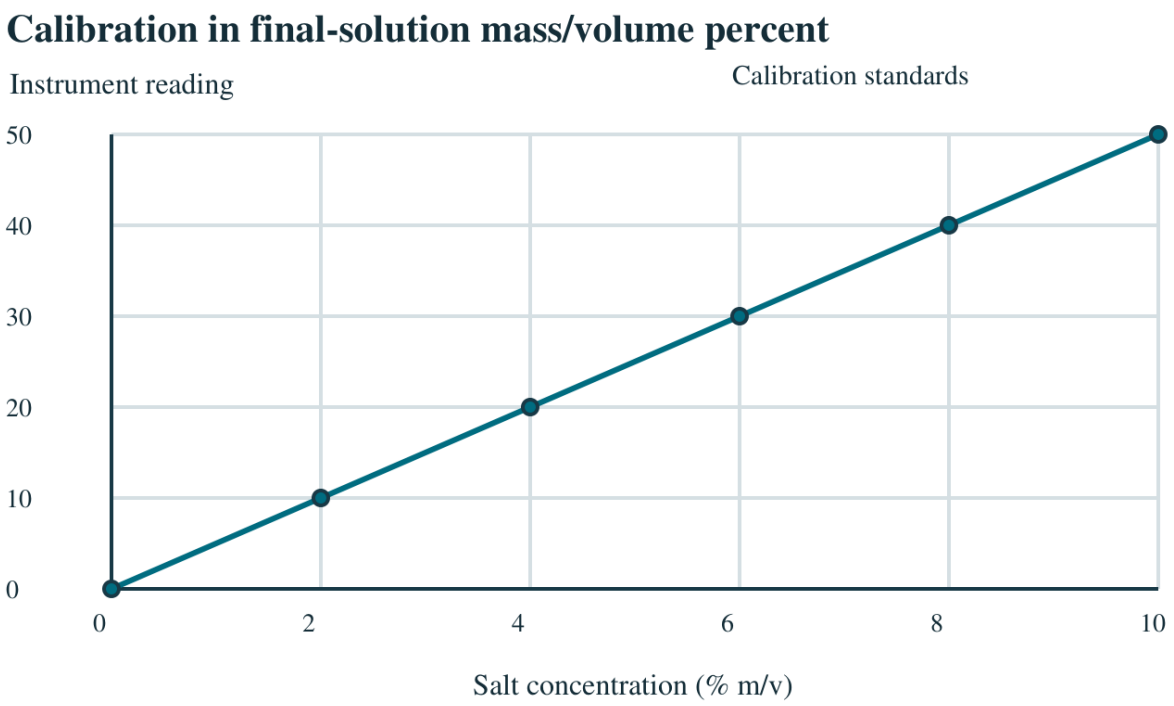
Answer: _____
_____
_____

29. (4 points) Use Figure F3. Why would a noon-only survey miss information visible in this graph?

Refer to F3. Diel dissolved oxygen (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. A sample gives reading 22. Interpolate its salt concentration.



F4. Salinometer calibration (original synthetic teaching figure)

Answer: _____
_____
_____

31. (1 points) Use Figure F4. How many grams of salt are needed for 200 mL final solution at the 4% standard?

Refer to F4. Salinometer calibration (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (1 points) Use Figure F4. Why is extrapolating this line to a reading of 80 less supported than interpolation?

Refer to F4. Salinometer calibration (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____